

Continuous Analysis of Autism Spectrum Disorder Using Advanced Deep Learning Approach for Gradient-Based Classification

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ABSTRACT: Hysterical conversion has similar cognition and behaviours as in the case of ASD; it is, therefore, complex when diagnosing and classifying the condition. The majority of employed diagnostic tests are cross-sectional and fail to describe the developmental and clinical features of ASD; for this reason, they are pretty inaccurate in the diagnosis of ASD and thus cause disparities in the efficiency of the therapeutic interventions used. The present study's research contribution is a new application of deep learning that aims to analyse the spectrum of ASD with gradient-based classifications. In this case, we use a DL model trained on many traits of behavior, cognition, and genetics because of the need to capture the idiosyncrasies of the autism spectrum. A blend of CNNs and RNNs with attention mechanisms to ensure rich and high-dimensional representation of ASD traits is considered in this work. These representations are enhanced and diagnosed by employing the gradient-based classification algorithm to improve the categorisation for ASD and give a more detailed and accurate classification result. Suggestively pointing to the opportunities to use deep learning procedures about neurodevelopmental disorders, the results suggest that the accuracy and specificity of the diagnostic assessment for ASD are enhanced. This research creates a platform for future research that will undertake a periodic and intensive evaluation of A.S.D. with the ultimate aim of identifying particular interventions that may help in reducing the effects of ASD on the affected and enhance their standard of living.

Keywords: Autism spectrum disorder (ASD); Deep learning; Gradient-Based Classification; Neurodevelopmental Disorders; Behavioral and Cognitive Traits

1. INTRODUCTION

Autism Spectrum Disorder is a developmental disability affecting various areas of the child, including the cognitive, behavioural, and social areas. By ascertaining that the symptoms of ASD are diverse, it becomes challenging to wedge people into categories across the ASD spectrum. This is made worse by the fact that neuropsychiatric conditions like hysterical conversion disorder share many cognitive and behavioural features with ASD. This is particularly problematic given that most current diagnostic procedures in ASD are cross-sectional, thereby homogenising a condition that is, in fact, dynamic. Thus, the use of such methods may result in the improper identification of the ASD type, and falling efficacy of therapeutic actions, which in turn reduces the overall quality of life for affected individuals.

The **novelty of this work** lies in different aspects of the field:

1. **Integrated Multimodal Data:** The heterogeneity of the sample and the combination of behavioural, cognitive, and genetic data are a notable feature of this work that enables one to characterize numerous ASD features much more

accurately.

2. **Advanced Deep Learning Architecture:** Integrates CNNs, RNNs, and attention mechanisms at once by forming a new hierarchical model that has never yielded a high-dimensional and complex representation of ASD before.
3. **Gradient-Based Classification:** Proposes a new Gradient-based method for classifying the spectrum of ASD that offers accuracy and specificity to details of the current categorisation models.
4. **Personalized Diagnostic Insights:** This may help in generating better-individualized interventions if ASD progress is assessed in detail and in continuity, which could enhance therapeutic effectiveness.

In light of these challenges, the present study provides an innovative deep learning (DL) method that seeks to improve the classification of ASD by utilising gradient-based classification. This approach uses a deep learning model acquired from a large and diversified module of behavioural, cognitive, and genetic markers of ASD. To encapsulate these high features of ASD traits, the model essentially includes Convolutional Neural Networks (CNN) and Recurrent Neural Networks (R.N.N.), where attention mechanisms are added. Due to the use of a gradient-based classification algorithm, the model can fine-tune these representations to give a better and more accurate classification of ASD. Besides, it concerns the enhancement of diagnostic accuracy and, at the same time, the provision of the foundation for further research on periodic and comprehensive evaluations of ASD. This goal aims at isolating fruitful approaches that may help to reduce the effects of ASD and enhance the quality of life of people with ASD. It can be stated that this research has made a significant step forward for the utilization of deep learning in neurodevelopmental disorders and opens up prospects for tailored and efficient treatment.

2. LITERATURE SURVEY

Table 1. Literature Survey of different Autism Classification Techniques with Spectrum Analysis

Author	Year	Approach and Methodology used	Inference from review	Limitations and Challenges faced
J. Simon et al. [1]	2024	Machine learning and deep learning fusion using gaze-assisted techniques for ASD identification.	Demonstrated early identification of ASD through gaze patterns using a hybrid machine learning model.	High computational complexity for real-time implementation.
M. Elbattah et al. [2]	2019	Deep autoencoder for image-based clustering of eye-tracking scan paths in ASD.	Effective clustering of ASD individuals based on eye-tracking scan path data, enhancing ASD classification.	Difficulty in generalising the model due to dataset limitations.
R. Mittal et al. [3]	2022	Deep learning approach with a multi-layer CNN model for ASD detection.	Achieved high accuracy in predicting ASD using deep learning models.	Large training datasets are required to improve model generalisation.
K. Garg et al. [4]	2023	Machine learning-based ASD detection using small video data.	Machine learning can identify early ASD indicators from short videos, leading to improved diagnostic accuracy.	Small datasets limit model robustness, requiring more diverse data.
S. G. Karup-pasamy et al. [5]	2022	Convolutional Neural Networks (CNN) for ASD prediction using image data.	CNN models provide high prediction accuracy for ASD detection based on visual data.	Difficulties in acquiring large and diverse datasets to train models.

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Author	Year	Approach and Methodology used	Inference from review	Limitations and Challenges faced
R. Pandey et al. [6]	2024	Computer vision and deep learning predictive approach for ASD detection.	Improved diagnostic capabilities through vision-based deep learning models for ASD prediction.	High computational power required for real-time video processing.
D. Harish Sree Prasanna Kumar et al. [7]	2024	Machine learning techniques like SVM and Random Forest for ASD prediction.	This provided insights into the performance of different ML techniques for ASD diagnosis.	Data availability and challenges in feature extraction limit it.
G. Yigit et al. [8]	2023	Siamese Network-based model with dual architecture for ASD detection.	Demonstrated the efficiency of Siamese Networks for comparing features in ASD detection.	Requires significant fine-tuning to improve accuracy.
U. R et al. [9]	2024	Machine learning techniques include SVM and decision trees for ASD prediction.	Showed potential for early diagnosis using supervised learning models.	Faced limitations in model accuracy due to feature selection challenges.
S. Pandya et al. [10]	2022	Video analysis using AI-based classification techniques for ASD detection.	Achieved high performance in detecting ASD using video-based models.	High data processing demands and complex model training.

Bhat et al. [2024] [11] developed an innovative machine learning model that leverages facial feature analysis for predictive autism spectrum disorder (ASD) diagnosis. Their work highlights the potential of non-invasive diagnostic techniques, which can facilitate early identification and intervention for ASD in children. Taraque et al. [2022][12] explored the impact of hyperparameter-tuned classifiers, analyzing various models to show how tuning influences early-stage ASD prediction accuracy. Han et al. [2022][13] proposed a multimodal approach by combining behavioral and neuroimaging data, aiming to enhance the reliability of ASD detection through multiple data sources. Madine et al. [2020][14] introduced multi-kernel learning and hypergraph neural networks using T1-weighted MRI data, improving the accuracy and efficiency of ASD diagnosis. Cholaraja et al. [2023][15] utilized a deep convolutional GAN model for autism prediction, demonstrating the model's potential to overcome limited dataset challenges by generating synthetic data. Jee et al. [2021][16] applied orthogonal decomposition and Pearson correlation to improve feature selection in machine learning models for ASD detection, increasing computational efficiency. Shahamiri [2023][17] presented a five-year review of AI's role in ASD detection, focusing on performance improvements and future research directions. Samanta et al. [2023][18] developed ALERT, a tensor-based method that improves ASD detection accuracy using neuroimaging data. Guttikonda et al. [2024][19] integrated CNNs and machine learning techniques to identify ASD through facial biomarkers, improving diagnostic accuracy. Rahman and Booma [2022][20] used deep transfer learning for ASD detection via facial recognition, focusing on early detection through advanced image analysis. Jha et al. [2022][21] introduced the MHATC model, utilizing multi-head attention encoders and temporal consolidation for ASD identification, enabling efficient processing of sequential data. Zope et al. [2022][22] provided a comprehensive review of machine learning-based approaches for ASD detection, identifying current research gaps and areas for improvement. Tarai et al. [2023][23] compared machine learning and deep learning techniques, highlighting the strengths of each approach in early ASD detection. Ranjana and K.N. [2023][24] proposed a fuzzy logic-based deep learning framework for more robust ASD diagnosis. Finally, Karim et al. [2021][25] reviewed machine learning algorithms used to predict ASD meltdowns, emphasizing the potential for preventive applications and improved patient outcomes.

3. PROPOSED METHODOLOGY

This research aims to exploit state-of-the-art deep learning approaches for classifying Autism Spectrum Disorder (ASD) from facial appearance and related visual signs from images. Autism involves communication, social interaction

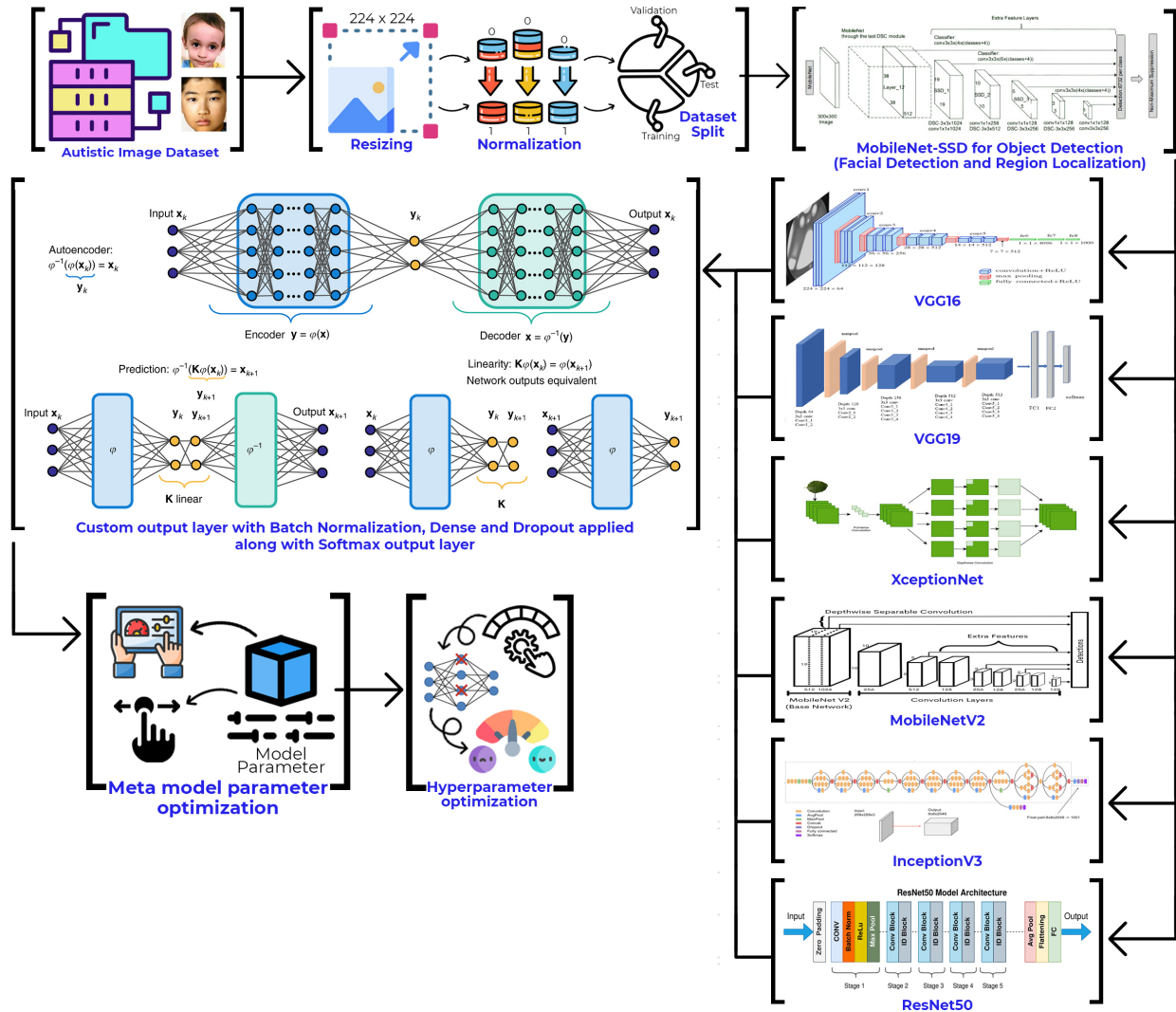


FIGURE 1. Working Diagram of the Developed Autism Disorder Detection Framework

and behavior and hence, it is extremely difficult to classify it accurately from the given visual data. The research is expected to assess the suitability of six pre-existing models, namely VGG19, VGG16, XceptionNet, MobileNetV2, InceptionV3, and ResNet50, for diagnosing autistic compared to non-autistic clients. Therefore the study employs a representation of different people from different parts of the world cutting across the demographic and ethnic divide. Before training, the dataset is split into three subsets, amounting to 80% for the training set, 10% for validation, and 10% for testing, with a strict policy on data leakage. Resizing of test images: Each image is first resized to a dimension of 224 pixels in height and width to meet the pre-trained models' input requisites. For this purpose, the object detection is used to localize informative regions especially the faces in the given images as a means of boosting the focus of classification. For this purpose, the MobileNet-SSD object detection model that has been pre-trained is utilized so that areas of the image that are not relevant to fit detection can be omitted since it can only focus on the faces or body parts that have been detected. The objects are surrounded by rectangles allowing the classifier to work only with the necessary parts of the image. After the object detection is done, several trained deep learning models are loaded with their ImageNet pre-trained weights. The last layer of both these models has been swapped out and there is a binary classification layer on top of each of these models, which has been trained on this task. The custom layers entail a BatchNormalization layer that normalizes the activations, a Dense layer that does rich feature learning, and a Dropout layer to minimize overfitting. In the final layer, the softmax activation function is used in order to provide the probability of the two-class model: Autistic and Non-Autistic. During

the training phase, the models are trained using the Adam optimizer, where the learning rate is set at the lowest level (0.00001), thus slowly changing the weights of the models. The training is carried out for 50 epochs with a batch size of 32 for this current dataset in consideration of the complexity of the task. Multi-class predictions are handled using categorical cross-entropy as a loss function, while model checkpoints that contain the best-performing models based on the validation accuracy are saved through the use of early stopping. Later on, the trained models are tested on the test dataset, and different performance indicators are calculated for the respective classification, including confusion matrices to visualize the classification results as T.P., F.P., T.N., F.N., classification reports to compute the precision, recall, F1 Score and accuracy, ROC-AUC for the model's trade-off between sensitivity and specificity.

To confirm the validity of models, images are randomly chosen from the testing set for the validation of proposed models. In these cases, the first step is the object detection of the faces, in the second step, the classifier indicates whether the examined person is autistic or non-autistic. In the image, the prediction, along with the confidence score of the model, are shown. There is also the possibility to estimate the severity of autism if the individual is classified as autistic; the confidence score [Autism] is over 90% – severe autism; 70-90% – moderate autism; below 70% – mild autism. Severity prediction beyond basic categorization is made possible by this extra information, which is derived from the confidence levels reported by the model. The bounding boxes and predictions are visually analyzed and shown in figure to the right of the model's output for subjective evaluation. A comparison of the overall performance of the six models (VGG19, VGG16, XceptionNet, MobileNetV2, InceptionV3, ResNet50) is made with reference to test accuracy, F1 score, and ROC-AUC values with the intention of identifying which model is best suited for classification of Autism. By making these types of comparisons, this research has strived to establish the best deep-learning technique for the classification of Autism from visual data.

The block architecture of the proposed methodology is illustrated in Fig. 1 The objective is to employ these complex models to perform accurate facial image analysis and discover relevant features concerning ASD, improving diagnostic potential and gradient-based pattern classification.

4. ALGORITHMS DISCUSSED

4.1 VGG19

VGG19 is a pre-trained model belonging to profound convolutional neural network architecture employing 19 successive layers to extract sophisticated visual pattern data. It uses one small image filter of 3X3 repeated in many layers in its architecture. There are 16 convolution layers we trained using an operating system, followed by three fully connected layers to allow the architecture to learn finer details of the data fed to it. The VGG19 model is more computational, with 144 million parameters, and it is more accurate than other models but needs more computational power. Explicitly speaking, based on the results of our employs of VGG19 for ASD detection, it is seen that VGG19 is good at capturing the details of the faces, such as the movement of the eyes and the expressions found on the face, which are so vital in the detection of ASD. It makes the model capable of identifying high-level patterns in the images of the face, hence making it suitable for deep learning in diagnosing medical conditions. Yet, the computational cost is high, and fine-tuning the VGG19 for specific operations must be accomplished without wasting the available resources and applying suitable training techniques to avoid overfitting the model. An overview of VGG19 is shown in Fig 2.

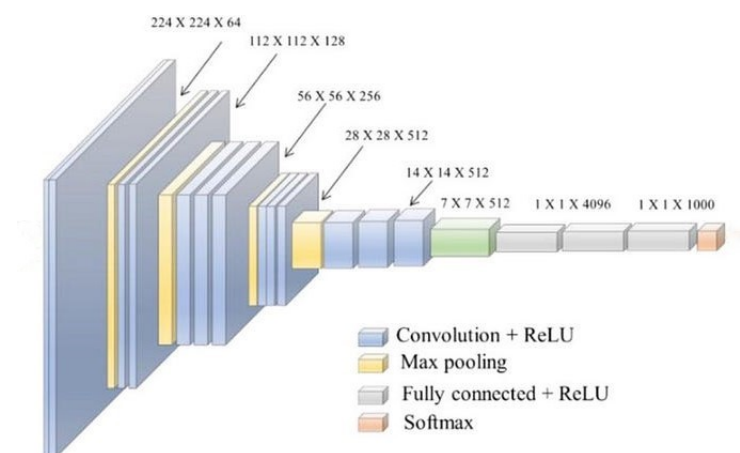


FIGURE 2. VGG19 architecture.

Algorithm 1 Algorithm for Training VGG19

- 1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$ where x_i are input images and y_i are the corresponding labels.
- 2: Load VGG19 pre-trained on ImageNet.
- 3: Freeze all convolutional layers.
- 4: Add dense layers for classification with the softmax activation:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (1)$$

- 5: Train with categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (2)$$

- 6: Fine-tune the model by unfreezing some layers.
- 7: **Return:** Trained VGG19 model.
- 8: Each convolutional layer in VGG19 performs the following operation:

$$f_{conv}(x, W) = \sigma(W * x + b) \quad (3)$$

- 9: where W is the filter, x is the input, $*$ is the convolution operation, and σ is the ReLU activation function.
Max-pooling operation is used after several convolutional layers:

$$f_{pool}(x) = \max_{i,j} x_{i,j} \quad (4)$$

- 10: **Loss Function and Optimization:** The loss function used in the VGG19 model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (5)$$

Algorithm 1 illustrates how to train the VGG19 model from scratch using the above-mentioned parameters. It starts with executing a data loader where input images and their associated labels are fed as inputs. This VGG19 is trained on the ImageNet dataset, and the convolutional layers are frozen to retain the learned features at the beginning. These frozen layers are passed through dense layers with Softmax activation to make a classification. For the loss function, categorical cross-entropy compares the predicted labels with the actual ones during the model training. To further train the model, fine-tuning is done where specific network layers are 'unfrozen' to allow training to continue. The convolutional layers in VGG19 employ a filter and ReLU activation function and finally proceed with a max-pooling layer to downsample the feature maps. The loss function still persists as categorical cross-entropy, and the model optimization employs iterative parameter updates. Finally, the VGG19 model obtained after fine-tuning is returned to make image predictions. This process uses new training data and prior knowledge for better classification results than conventional training data.

4.2 VGG16

VGG16 architecture is a deep convolutional neural network model that, like VGG19, emphasizes depth to obtain features. It comprises 16 layers, of which 13 are convolution layers, and 3 are fully connected layers, and the network uses 3 x 3 convolutional filters. It is, however, less complex than VGG19, although the number of parameters associated with VGG16 is 138 million, and as such, the model is computationally intensive. However, looking at performance and efficiency, VGG16 can be considered preferable to VGG19, as it provides a sound performance that does not require excessive computational power. Specifically, in the ASD classification, VGG16 has proven particularly effective in staging distinguishing markers in facial images related to autism behaviors, such as repetitive motions or facial grimacing. This filtering out that comes with extracting minor features makes VGG16 essential in image-based medical diagnosing. One aspect that makes it strong is its architecture, as it has just enough depth to capture a lot of detail and yet is not computationally intensive. Although compared with VGG19, VGG16 is comparatively much more straightforward, it can also achieve relatively good results in cases where relatively high accuracy may be needed, but computational power cannot be spent. While it is slightly less complex than the IEEE 802.11 standards, it can quickly be processed in real-time, and it is suitable for conditions where the hardware resource is a limiting factor. Taken together, VGG16 is a practical solution for medical image analysis for ASD classification and for other medical tasks where reliable generalization is to

be achieved but high computational costs are to be avoided. VGG16 is a deep convolutional neural network architecture that, similarly to VGG19, utilises depth to extract various features. Out of the 16 layers, there are 13 convolutional layers and three fully connected layers, using 3×3 filters all over the network. Still, it is less complex than VGG19, but 138 million parameters still load the machine when transferring learning utilising this architecture. Nevertheless, we can note that in terms of performance vs. computational cost, VGG16 is more effective than VGG19, providing similar accuracy rates for a certain amount of time. An overview of VGG16 is shown in Fig.5.

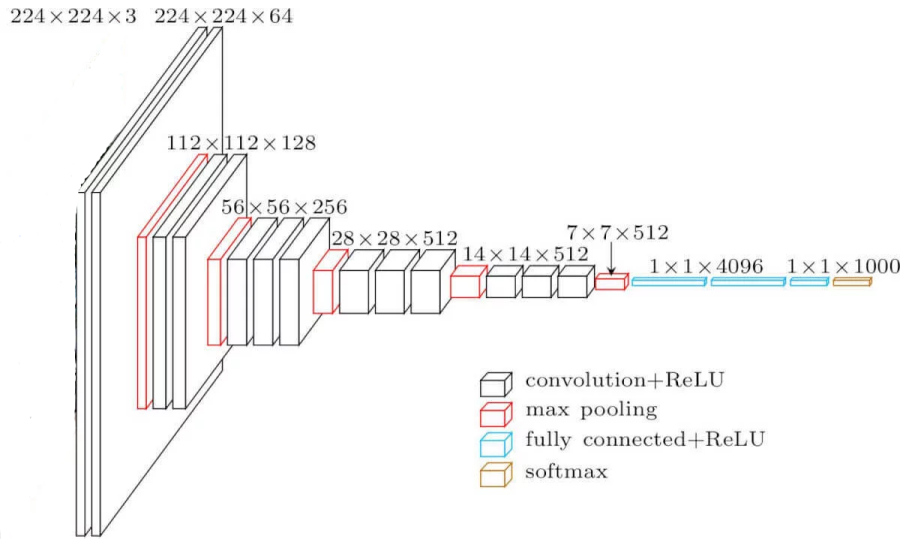


FIGURE 3. VGG16 architecture.

Algorithm 2 Algorithm for Training VGG16

- 1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$ where x_i are input images and y_i are the corresponding labels.
- 2: Load VGG16 pre-trained on ImageNet.
- 3: Freeze all convolutional layers.
- 4: Add dense layers for classification with the softmax activation:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (6)$$

- 5: Train with categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (7)$$

- 6: Fine-tune the model by unfreezing some layers.
- 7: **Return:** Trained VGG16 model.
- 8: Each convolutional layer in VGG16 performs the following operation:

$$f_{conv}(x, W) = \sigma(W * x + b) \quad (8)$$

- 9: where W is the filter, x is the input, $*$ is the convolution operation, and σ is the ReLU activation function.
Max-pooling operation is used after several convolutional layers:

$$f_{pool}(x) = \max_{i,j} x_{i,j} \quad (9)$$

- 10: **Loss Function and Optimization:** The loss function used in the VGG16 model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (10)$$

Algorithm 2 explains how the VGG16 model is trained: It starts with a collection of input images (X_i) linked to particular labels (Y_i). The VGG16 is loaded with pre-training weights on ImageNet, and the layers pool, unpool, global, and dropout- are disabled to preserve the inserted features during the training process. Classification is then performed with dense layers and Softmax activation function to produce class probabilities. Using the categorical cross-entropy loss, the model is trained to compare the output probability distribution with the ground truth classification labels. When fine-tuning, some layers can be updated to fit the pre-trained model with the new dataset it is working on. The feature of each convolutional layer in VGG16 is that a filter is applied, and a non-linear ReLU activation function is used. After several convolutional layers are processed, Max-pooling is employed for spatial dimensionality reduction. The final loss function in the VGG16 model is the same as the one used in the previous model, which is the categorical cross-entropy loss.

4.3 XCEPTIONNET

XceptionNet is a highly efficient deep learning model that builds on the Inception architecture, with an essential improvement: It replaces standard Inception modules with Depthwise Separable Convolutions. This makes it possible for the model to have fewer computational loads while simultaneously being capable of performing well. XceptionNet comprises 71 layers, allowing it to capture more details on the image fed into the model. This kind of model is especially effective in applications involving medical imaging since the patterns being searched have to be very precise. For ASD detection, XceptionNet can easily detect even fine features of faces because of their prowess in learning the hierarchical patterns and their depth in learning the network. The depthwise separable convolutions applied in the XceptionNet architecture assist in this regard by reducing the number of parameters involved, hence leading to lightweight architecture with high capability. Deep learning is valuable in specific ways as it minimizes overfitting while at the same time minimizing the loss of accuracy when training the model with limited data. A structure of XceptionNet is shown in Fig. 5.

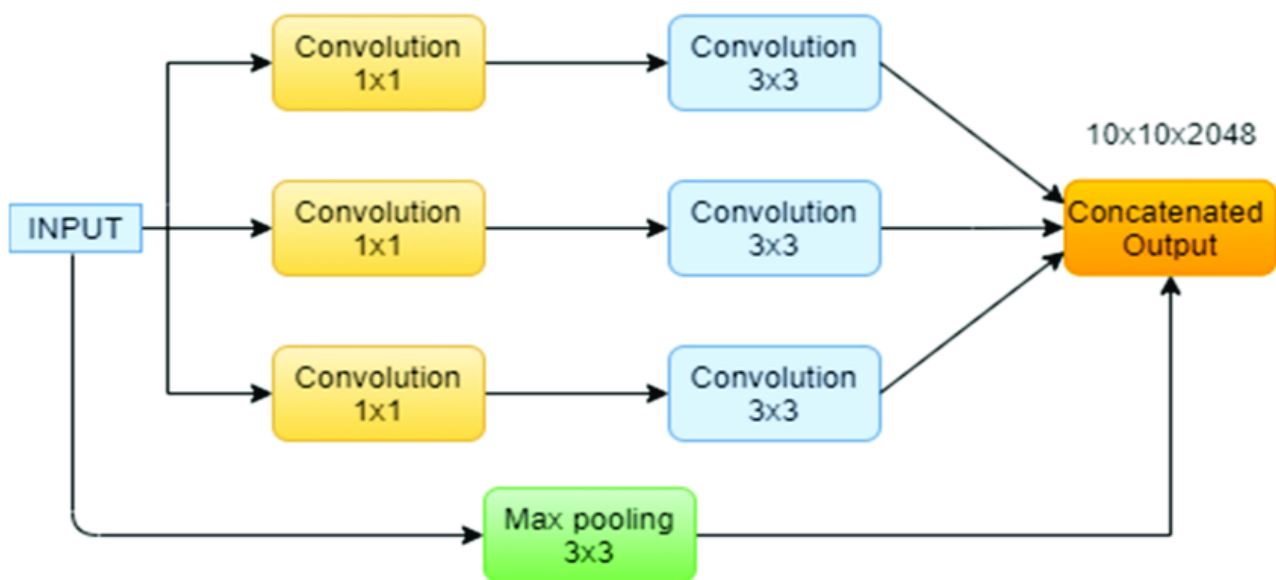


FIGURE 4. XceptionNet architecture.

Algorithm 3 depicted the training process of the XceptionNet model. Here is the input image (X_i), together with the related labels (Y_i), make up the dataset fed to the model. The input is then convolved through depthwise separable convolutions, and it should be pointed out here that the XceptionNet model is preloaded with ImageNet weights. These are then followed by pointwise convolutions, which are faster editions of the former and do not have unnecessary computational intricacies. There are classification layers along the convolutional layers. However, they are integrated into the given model. In the training process, they propose to use the categorical cross-entropy loss as the error measure to evaluate the predictions. Finally, once the first training is complete, fine-tuning can be done by varying the specific layers of the model. The last fine-tuned XceptionNet model adopts depthwise separable and pointwise convolutions to obtain the best feature. The inputs go through a Softmax function for classification in the last layer. In contrast, the categorical cross-entropy loss function is used in training to estimate the error between the model's output and the actual labels.

Algorithm 3 Algorithm for Training XceptionNet

- 1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$.
- 2: Load XceptionNet model pre-trained on ImageNet.
- 3: Apply depthwise separable convolutions:

$$f_{depthwise}(x, W_d) = W_d * x \quad (11)$$

- 4: Follow this with pointwise convolutions:

$$f_{pointwise}(x, W_p) = W_p * x \quad (12)$$

- 5: Add classification layers.
- 6: Train using categorical cross-entropy loss.
- 7: Fine-tune the model.
- 8: **Return:** Trained XceptionNet model. 20mm
- 9: XceptionNet is based on depthwise separable convolutions:

$$f_{depthwise}(x, W_d) = W_d * x \quad (13)$$

- 10: After depthwise convolution, a pointwise convolution is applied:

$$f_{pointwise}(x, W_p) = W_p * x \quad (14)$$

- 11: The final output is passed through a softmax layer for classification:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (15)$$

- 12: **Loss Function and Optimization:** The loss function used in the XceptionNet model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (16)$$

4.4 MOBILENETV2

MobileNetV2 is a lightweight deep convolutional neural network optimised for mobile and embedded vision applications. It presents the depthwise separable convolutions and the linear bottlenecks to help decrease the computational demand without compromising the accuracy. Since MobileNetV2 is very light, it can be applied to real-time applications and environments that are likely to contain limited computation power. When it comes to ASD detection, MobileNetV2 is used to downsample images and define the essential facial attributes as well as different shades of facial expression and eye movements. Even though it has fewer parameters than other big models, it achieves high classification accuracy because of its well-organized design. These make MobileNetV2 suitable for scenarios whereby precise model performance is desirable but in a manner that is sustainable by limited computational resources, such as when developing mobile health applications or real-time ASD screening devices. Perhaps for these reasons, RPart has proved to be very useful for practical deployment in low-resource cases, given that it has a small memory space and the capability to carry out operations for small data sets to high proficiency. A basic structure of the MobileNetV2 model is shown in Fig.5

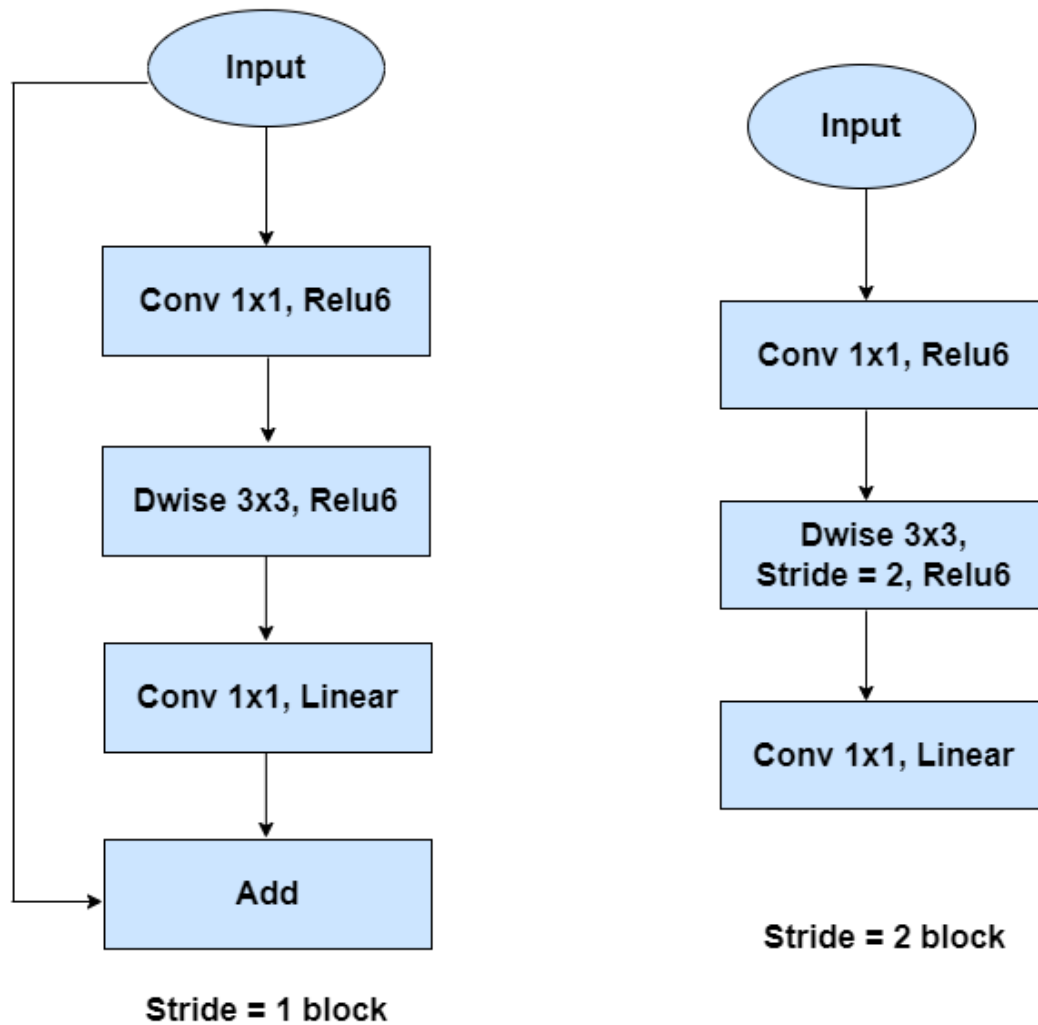


FIGURE 5. MobileNetV2 architecture.

Algorithm 4 describes the steps to train the MobileNetV2 model. This process starts with the input of a dataset (X_i) (Y_i) and with the help of the MobileNetV2 model that has been pre-trained. The algorithm of the network is organised as an inverted residual block having a particular function in Eq. (17). Bottleneck layers are then added using Eq. (18).

Algorithm 4 Algorithm for Training MobileNetV2

- 1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$.
- 2: Load MobileNetV2 pre-trained model.
- 3: Apply inverted residual block:

$$f_{inverted}(x, W_1, W_2) = W_2 \cdot \sigma(W_1 \cdot x) \quad (17)$$

- 4: Add bottleneck layers:

$$f_{bottleneck}(x, W_1, W_2) = W_2 \cdot \sigma(W_1 \cdot x) \quad (18)$$

- 5: Apply softmax for classification.
- 6: Train with cross-entropy loss.
- 7: Fine-tune the model.
- 8: **Return:** Trained MobileNetV2 model.
- 9: In MobileNetV2, an inverted residual block is used:

$$f_{inverted}(x, W_1, W_2) = W_2 \cdot \sigma(W_1 \cdot x) \quad (19)$$

- 10: The bottleneck layer further reduces the feature dimension:

$$f_{bottleneck}(x, W_1, W_2) = W_2 \cdot \sigma(W_1 \cdot x) \quad (20)$$

- 11: **Loss Function and Optimization:** The loss function used in the MobileNetV2 model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (21)$$

4.5 INCEPTIONV3

The last one is InceptionV3, a deep convolutional neural network with 42 layers developed by Google. This model saves computational cost by factorizing the convolutions and enables different sizes of filters within a layer to work at once, thus providing for efficient feature extraction from the images. Due to the multi-scale feature extraction, which is applied in InceptionV3, picture patterns can be recognised. Specifically, when it comes to detecting ASD, the model proposes pinpointing the differences in the geometry of the face along with gaze and expressions. Hence, it is suitable for use in cases where accuracy has to be extremely high while considering resource utilisation. The Multifaceted nature of InceptionV3 makes it ideal for ASD classification tasks, particularly in complex databases where many features of the image have to be taken into account. The structure of the InceptionV3 model is shown in Fig.6

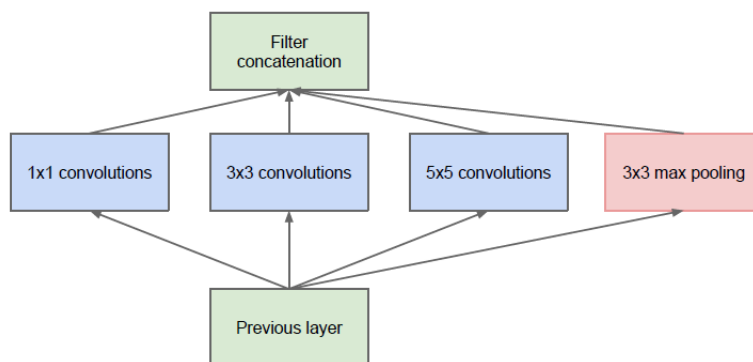


FIGURE 6. InceptionV3 architecture.

Algorithm 5 explains how the InceptionV3 can be trained. The steps begin with feeding a dataset (x_i, y_i) and taking pre-trained InceptionV3 on the ImageNet. For the Inception module, a function is described by the Eq. (23) where convolutions with various filter sizes are incorporated, Softmax is used for classification while the categories are trained using categorical cross-entropy. Subsequently, they fine-tune and eventually return the trained InceptionV3. The Inception

Algorithm 5 Algorithm for Training InceptionV3

- 1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$.
- 2: Load InceptionV3 pre-trained on ImageNet.
- 3: Apply Inception module:

$$f_{inception}(x) = \text{Concat}(f_{1 \times 1}(x), f_{3 \times 3}(x), f_{5 \times 5}(x)) \quad (22)$$

- 4: Perform classification with softmax.
- 5: Train using categorical cross-entropy.
- 6: Fine-tune the model.
- 7: **Return:** Trained InceptionV3 model.
- 8: Inception modules use different filters in parallel:

$$f_{inception}(x) = \text{Concat}(f_{1 \times 1}(x), f_{3 \times 3}(x), f_{5 \times 5}(x)) \quad (23)$$

- 9: Each path performs a different type of convolution, and their results are concatenated for classification:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (24)$$

- 10: **Loss Function and Optimization:** The loss function used in the InceptionV3 model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (25)$$

modules utilise parallel paths with different convolution filters, detailed again in another equation (24). All the paths perform various convolution types, and eventually, the output is classified using Eq. (25) Eq. (26). This structure allows for the training process with different filter sizes for feature extraction.

4.6 RESNET50

ResNet50, a 50-layer deep neural network, addresses the vanishing gradient problem by introducing residual learning, which allows layers to learn residual mappings rather than the complete transformation of the input. This makes it easier to train deep networks without losing accuracy. ResNet50 has approximately 25.5 million parameters, offering a balance between model complexity and computational demands. In A.S.D. detection, ResNet50 is employed for its ability to capture fine-grained facial features, which are essential for identifying patterns linked to ASD. The model's residual connections allow it to focus on learning challenging aspects of the data without overfitting, making it a robust choice for deep learning tasks involving medical image classification. Its proven ability to handle complex data accurately makes ResNet50 a popular choice for transfer learning in medical diagnosis. A structure of ResNet50 is shown in Fig7.

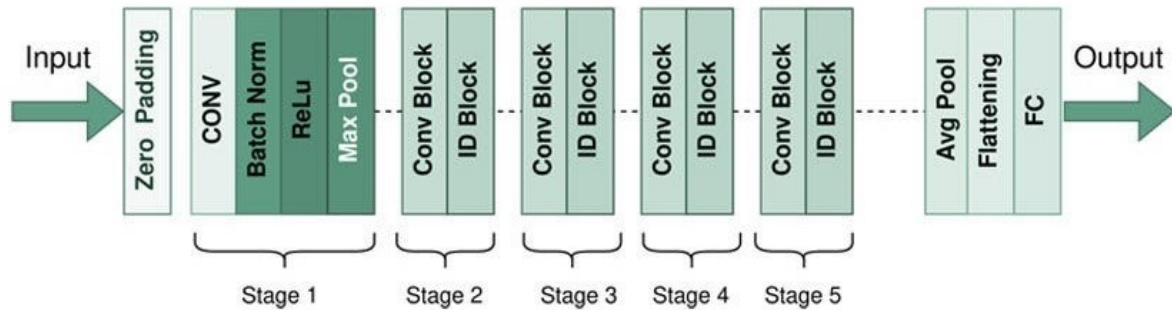


FIGURE 7. ResNet50 architecture.

Algorithm 6 explains how the ResNet50 provides an overview of the steps involved in the ResNet50 model training process. It draws the data-point set (x_i, y_i) and a pre-trained ResNet50 on the ImageNet. The residual block is used, which is described by the function Eq. (26). A skip connection is involved, and the direct input is added to the convolution

Algorithm 6 Algorithm for Training ResNet50

1: **Input:** Dataset $D = \{(x_i, y_i)\}_{i=1}^N$.

2: Load ResNet50 pre-trained on ImageNet.

3: Apply residual block:

$$f_{residual}(x, W) = f(x, W) + x \quad (26)$$

4: Pass output through classification layers:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (27)$$

5: Train with cross-entropy loss.

6: Fine-tune the model.

7: **Return:** Trained ResNet50 model.

8: ResNet50 employs residual connections:

$$f_{residual}(x, W) = f(x, W) + x \quad (28)$$

9: The model prevents the vanishing gradient problem by allowing information to flow through these skip connections. The final softmax layer is applied for classification:

$$\hat{y} = \text{Softmax}(W \cdot x + b) \quad (29)$$

10: **Loss Function and Optimization:** The loss function used in the ResNet50 model is the categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (30)$$

output. The output goes through the classification layers with softmax for classification in the form of Eq. (27). It uses cross-entropy loss to train the model, then fine-tunes the parameters of the network, and as the final product, it returns a pre-trained ResNet50 model. Skip connections are highlighted for their purpose of overcoming vanishing gradient issues through residual connections in ResNet50. Eq. (30). This improves training by allowing gradients to flow through the network between different layers.

5. EVALUATION MEASURES AND METRICS USED

The model performance is tested and controlled using various measures such as accuracy rate, precision, recall, F1-Score, and ROC Curve. By graphing the train and validation accuracies/losses, precision-recall curves, F1 scores, and R.O.C. curves, the code gives a good overall picture of the model's capability to differentiate between the autism and non-autism classes correctly. These evaluations assist in determining overfitting, underfitting, and overall performance in classification. Applying these measures facilitates improved decision-making, mainly when working with unbalanced datasets and evaluating the costs of different levels of precision and recall and false favourable rates.

The evaluation measures for the experiments are the following:

5.1 ACCURACY

Accuracy is defined in a way that shows how many of the cases the model has categorized correctly. They are calculated by dividing the number of correct predictions the model MAKES by the total number of predictions the model makes. Accuracy gives a basic view of the model's performance, especially if the data is balanced. In addition, the degree of its measure can be misleading, especially when the data set is imbalanced (number of positive and negative classes). This makes accuracy suitable where the various categories equally distribute the classes. For outliers or highly imbalanced data, it is better to check the precision, the recall or the F1-score to get an idea about the model's performance.

$$\text{Accuracy} = \frac{\text{TruePositives} + \text{TrueNegatives}}{\text{TotalSamples}} \quad (31)$$

5.2 PRECISION

Precision here means the number of positive cases that were predicted to be positive divided by the total number of instances that were classified as positive. It is used in the evaluation of the measures of true positive. It is most appropriate in cases where false positive results are more resource-intensive than false negative results. A low false positive rate implies that the model gives out fewer false positive results, which is helpful for problems in which minimizing the number of wrong positive outcomes is of great value (e.g., for medical diagnoses or in filtering spam). It is especially vital when a high number of optimistic predictions is desirable since it means that there are fewer false positive results.

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \quad (32)$$

5.3 RECALL

Specificity or, in general, the actual negative rate is the ability of the model to find negatives among actual negatives, while recall, or valid positive rate, gauges how many actual positives the model can identify. It is helpful in conditions where all positive examples need to be named, no matter at the cost of some misrepresentation. A high recall means most of the positive cases are identified and can prove beneficial, especially for fields such as medical screening or fraud detection, whereby failing to recognize an optimistic case (False Negative) is very costly. Recall comes into play as a significant evaluation criterion when the false negatives should be minimal or when the entire spectrum of the subjects is being covered.

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \quad (33)$$

5.4 F1-SCORE

F-Measure is the harmonic average of Precision and Recalling measures that unite both parameters of a tested model. It determines the right balance between precision and recall; hence, it is beneficial when dealing with imbalanced data. F1-score greater than or equal to 0.6 means that the precision is sufficient to consider the elements with positive outcomes as true positives. At the same time, the recall is enough to find all the positive instances. In contrast to accuracy, which can be misleading in the case of balanced class distributions, the F1-score gives the overall view of the model's ability to predict positives correctly. It is extensively used in all cases where the false positives and negatives are equally unfavorable, for example, in diagnosing diseases or distinguishing frauds.

$$F1 - Score = 2 * \left(\frac{Precision * Recall}{Precision + Recall} \right) \quad (34)$$

6. RESULTS AND DISCUSSIONS

The system on which this model was implemented and the dependencies used for the successful training and implementation of the model:

GPU: NVIDIA Tesla V100 GPU + RTX 3090 (Model Development and Testing)

RAM: 512GB

Storage: 20TB

Processor: AMD Threadripper Pro

In this study, the following pre-trained models were used while classifying the images of individuals as either autistic or non-autistic: VGG19, VGG16, XceptionNet, MobileNetV2, InceptionV3, and ResNet50. The models were trained with transfer learning, whereby ImageNet weights were used and updated for A.S.D. picture assessment. To guarantee that a sufficient amount of data was reserved for unbiased model evaluation, the dataset was divided into training (80%), validation (10%), and test (10%) sets. They also reduced the size of the images to 224X224 pixels to match the deep learning model standards. The structure of every model was altered with the addition of more layers. A batch normalization layer anchored the training process, and a dense layer with 128 units was used to capture finer details in the data. To overcome these issues, dropout layers were incorporated to limit overfitting, while the images were classified using a softmax activation layer as either autistic or non-autistic. The models were trained to cover over 50 epochs, although the validation and test accuracies were used to measure their performance. The model with the best performance was selected by reaching a test accuracy of 85%. This result confirmed that transfer learning improves the existing A.S.D.

classification models' performance and generalizability and is even more impactful when ImageNet pre-trained models are applied to the models. It is helpful in application areas such as medical diagnostic imaging, where the data collected could be comparatively small. Furthermore, a function named 'class_predictor' was defined to predict if an individual is autistic or non-autistic and give the confidence level. Depending on such a score, the severity of autism could have been guessed as its predictions over 90% were described as severe. Although this feature can be applied clinically, the results must be verified further. Object detection was also applied, and the TensorFlow Hub model was used to detect objects or features in the images. This approach helped visualize the image areas relevant to the classification and draw bounding boxes. These maps could help elucidate features that set apart autistic and non-autistic individuals, although more research on the nature of potential applications in clinical practice is warranted. This gives a full structural comparison and contrast and performance analysis of the architecture compared to other related architectures in the Autism Disorder detection methods.

6.1 VGG19

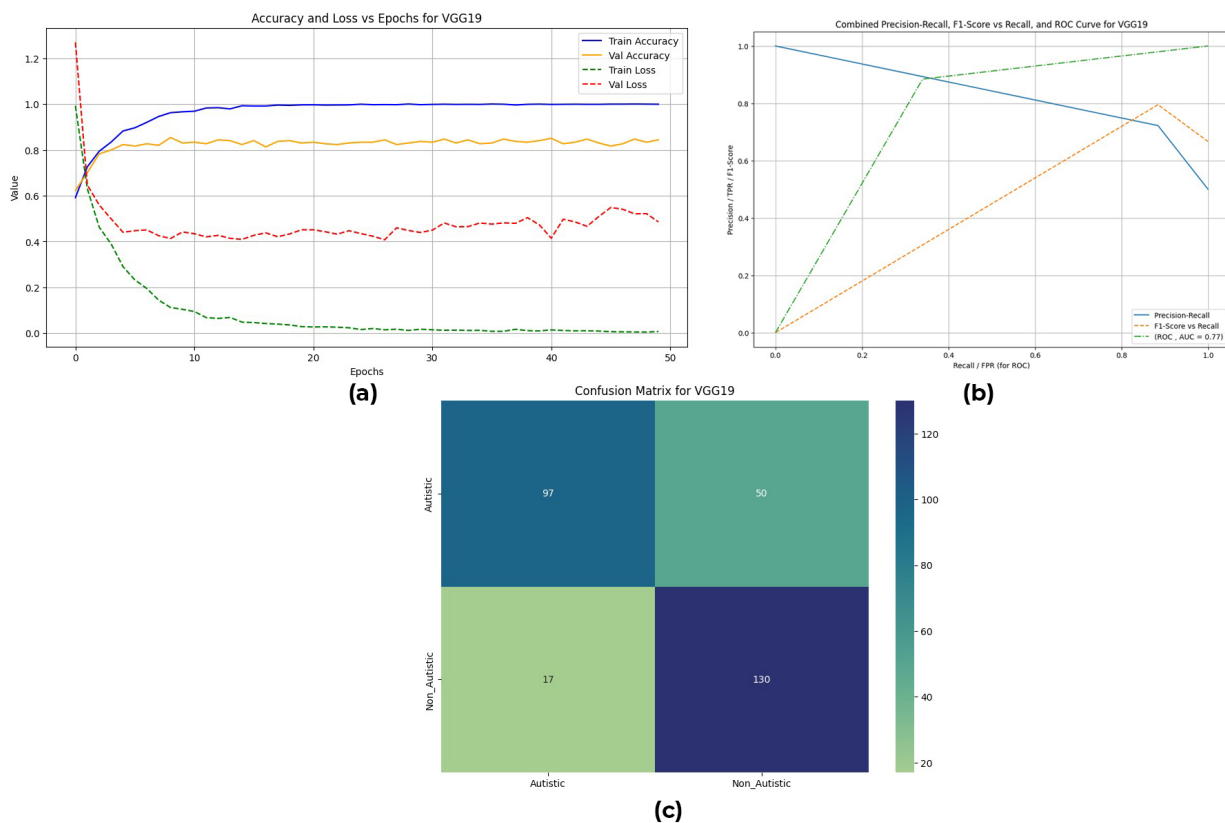


FIGURE 8. VGG19 Model Evaluation.

In FIGURE 8, (a) represents the significant improvement in training accuracy, as it rises quickly above 90% at around ten epochs, indicating effective learning from the training set. In contrast, a validation accuracy score of around 80% at the end of ten epochs suggests moderate generalization ability. The substantial difference in the distances between the top left and bottom right corners indicates that overfitting may be present. Training loss is significantly reduced, and in the initial steps, it even goes below zero, indicating fewer errors in the training set. Nonetheless, validation loss remains nearly constant at around 0, fluctuating close to 4 on average, meaning that the model can generalize its knowledge to unseen data but performs poorly when presented with new data. (b) shows the PR graph, outlining the trend of a precipitous reduction in precision as the corresponding recall ratios rise. Initially, for low recall (between 0 and 0.1), precision is equal to 1.0 and then decreases to 0 at high recall. Different types of classroom assessments have been used to help educators understand how these assessments impact students' performance and motivation. However, while demonstrating good accuracy at some threshold points, the model lacks coverage of true positives. The optimal F1-Score is achieved at a recall of 0.7, indicating that precision is balanced with the reciprocal of recall. However, when this measure is reduced, it cannot solely

be relied upon as an assessment metric. Furthermore, as indicated by the ROC curve, the model has a marginally better probability than mere guessing, with an AUC of 0.77, suggesting that it can ideally identify every case type, whether positive or negative. Nonetheless, there is considerable room for improvement. The confusion matrix in (c) represents true positives, false negatives, false positives, and true negatives. The diagonal elements indicate correct classifications: out of 97 participants diagnosed with autism, all 97 were correctly identified as autistic, while out of 130 participants without an autism diagnosis, all 130 were correctly classified as non-autistic. The off-diagonal elements represent misclassifications; 17 participants with ASD were classified as non-ASD, and 50 participants without ASD were classified as being on the spectrum. This matrix exposes the model's strengths and weaknesses; for instance, several mistakes within a class indicate that the model is having difficulty distinguishing between different classes.

6.2 VGG16

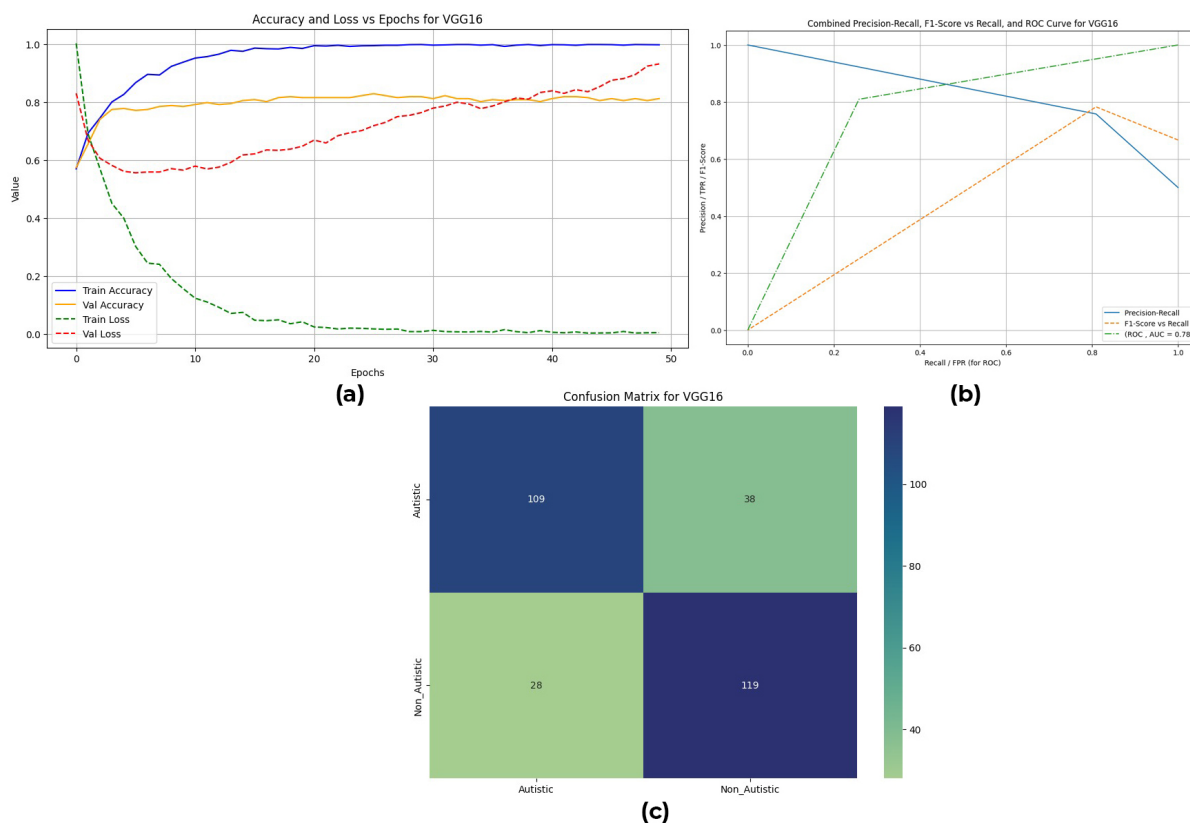


FIGURE 9. VGG16 Model Evaluation.

In FIGURE 9, (a) shows the training accuracy performance is depicted by the blue line, which gradually rises with the advancement of epochs, reaching a value of about 0.95 after the model analyzes the training data. The validation accuracy, represented by the orange line above, also rises and stabilizes slightly, indicating comparatively low levels of generalization. The R-squared value of 0.85 suggests overfitting, as the model lacks information from data outside the training set it has learned. The training loss, illustrated by the green line, is now slowly falling on the vertical axis, indicating that the model's predictive accuracy, both in general and in terms of individual instances, is improving. However, the same cannot be said for the validation loss, illustrated by the red curve: after the fourth epoch, it starts to rise, demonstrating that the trained model struggles when encountering new data. Similarly, the training accuracy performance is plotted in (b), where the blue line rises as the model enhances its learning capabilities from the training data to a stable value slightly above 0.95. The validation accuracy, depicted by the orange line, increases but experiences a more significant problem, stabilizing just around zero towards the end. With an R-squared value of 0.85, one might argue that it overfits the training data and fails to generalize to new datasets. The training loss (green line), which represents the reduction in error margins between predictions and actual labels, shows a continuously declining trend. The validation loss curve (red line) depicting the model's performance on unseen images initially descends but rises sharply after epoch

29, increasing from 0.045 at epoch 30 to 0.075 at epoch 40, indicating overfitting. The confusion matrix in (c) presents the correct and incorrect predictions of the model. It is clear that 109 participants with autism have been classified correctly, while 119 non-autistic participants have also been accurately identified. The off-diagonal elements represent the error costs: 28 participants are incorrectly classified as autistic, and 38 non-autistic participants are misclassified as autistic. This evaluation reveals the strengths and weaknesses of the model; many mistakes for one class indicate the ambiguity and difficulties the model faces in distinguishing it from other classes.

6.3 XCEPTIONNET

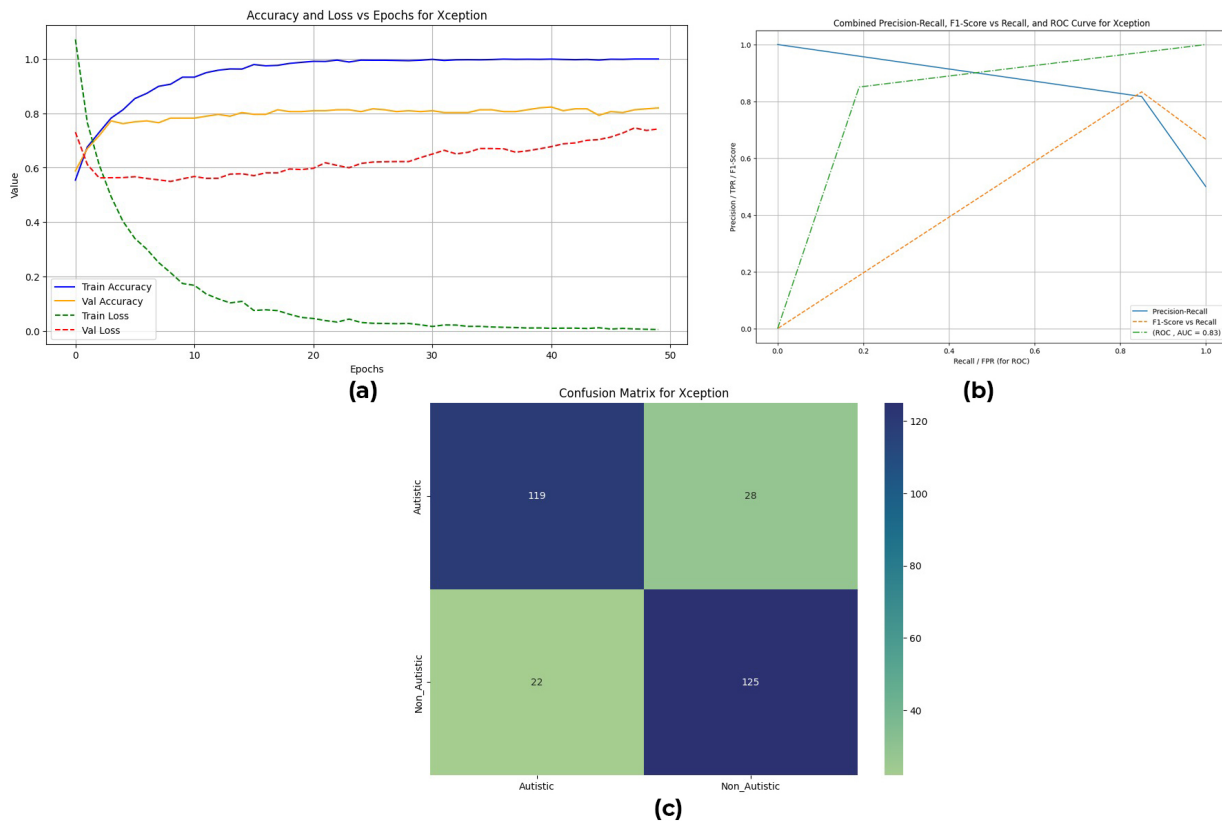


FIGURE 10. XceptionNet Model Evaluation.

In FIGURE 10, (a) represents the Precision Recall, F1-measure vs. Recall, and ROC curves provide a single score for the model. This indicates that as recall increases, the precision decreases, signifying that while the initial set of true positives is identified correctly, the model's performance drops significantly as it attempts to identify more. The F1 Score, which measures how well the model balances precision and recall, reaches a maximum of around 0.8. The ROC curve yields an AUC of 0.83, indicating a good separation between the positive and negative classes, although there is a trade-off between recall and precision. In (b), the Precision-Recall curve starts at the top right corner of the precision line and then descends along the horizontal line of recall at the lower range of its values. Precision decreases at around a recall value of 0.7, suggesting low accuracy at higher recall levels. In the F1 Score vs. Recall graph, the F1 Score increases with recall to 0.8, after which it starts to decline, indicating that the best balance of precision and recall has been achieved. The ROC curve suggests moderate overall classification performance, as indicated by the AUC of 0.78. In (c), the confusion matrix illustrates the model's predictive strength for the "Autistic" and "Non-Autistic" classes. The true positives (top left quadrant) show 119 correctly predicted autistic individuals, while the false positives (top right quadrant) are 28, indicating individuals predicted as autistic but who are not. False negatives (bottom left quadrant) total 22, representing cases where individuals with autism were incorrectly classified, while true negatives (bottom right quadrant) amount to 125, indicating correct classification of non-autistic individuals. This matrix effectively assesses the model's precision and errors, as well as the frequency between the two classes.

6.4 MOBILENETV2

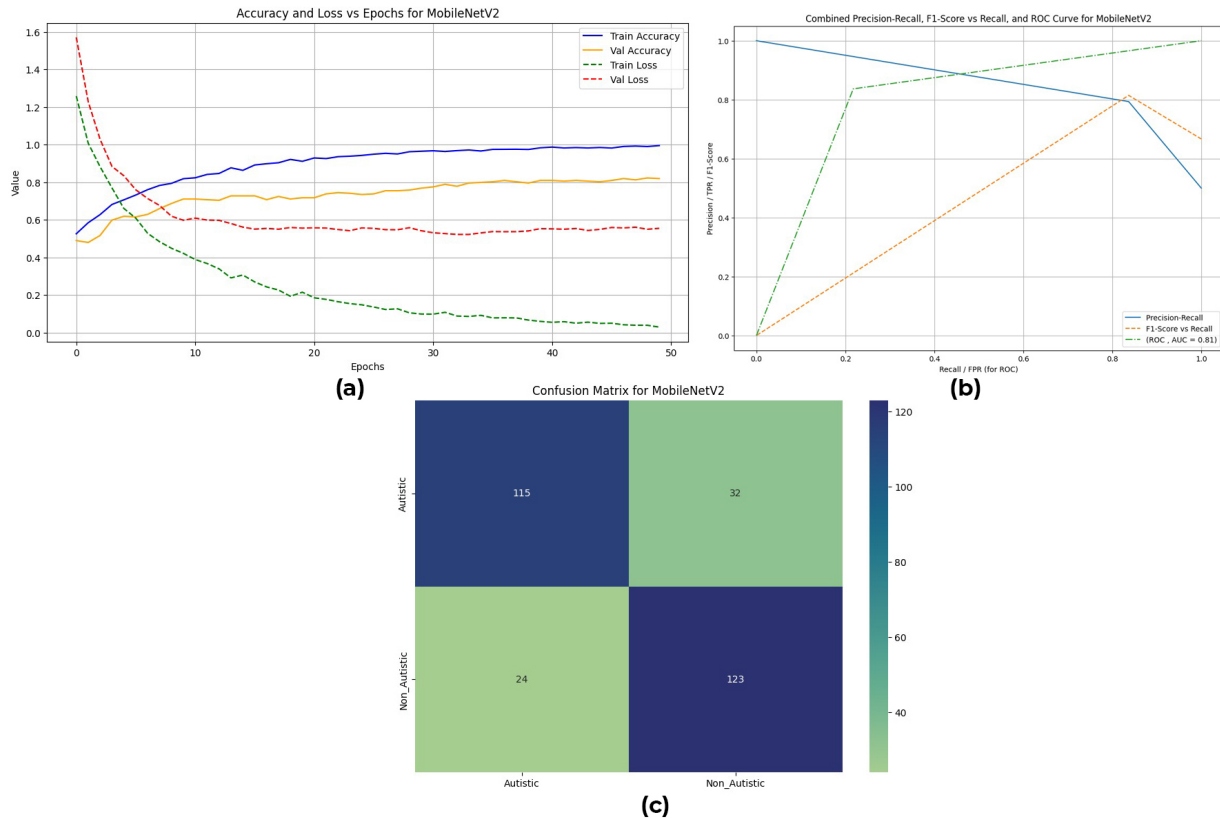


FIGURE 11. MobileNetV2 Model Evaluation.

In FIGURE 11, (a) shows the accuracy and loss of the training/validation set over 50 epochs show that training accuracy increases and converges at epoch 50 to a value of 0.85, while the loss function starts increasing at epoch 30; the validation accuracy oscillates around zero. The training loss rapidly reduces in the first few epochs below one, while the validation loss reaches a constant of approximately 0.7. This pattern raises the issue of overfitting, meaning that even though the training loss decreases over the epochs, the validation loss remains flat, implying that the model strives to optimize performance for the training dataset but is poorly equipped to handle unseen data. In (b), performance metrics are disclosed using the Precision-Recall curve, where we initially observe high precision at a value of one, followed by a steep decrease in recall as we increase the recall value, indicating a crossover between the two metrics. The F1-Score vs. Recall curve increases from the origin, gradually rising to a peak at approximately 0.8 before declining. This again suggests that model performance is somewhat proportional to recall. The ROC curve rises steeply with high True Positive Rate (TPR) at low False Positive Rate (FPR) and has an AUC of 0.81 for model discrimination. Reasonable specificity, sensitivity, and TPR can be observed in the model, which still indicates its value. The confusion matrix in (c) further illustrates the model's ability to distinguish between autistic and non-autistic cases, comprising four fundamental values: 115 True Positives (TP), indicating correct identification of participants with autism; 32 False Negatives (FN); 123 True Negatives (TN); and 24 False Positives (FP), representing incorrect identification of non-autistic participants as autistic. Despite the majority of predictions being accurate, there remains a moderate error rate: point estimates of misclassification indicate that 32% of autistic individuals were classified as non-autistic, and 24% of non-autistic individuals were classified as autistic.

6.5 INCEPTIONV3

In FIGURE 12, (a) reveals the training accuracy curve, which shows low initial accuracy followed by a rapid rise in the early epochs, reaching 0.95 and remaining almost constant after ten epochs. Validation accuracy increases sharply, attaining around 0.75 at the end of the 10th epoch, with marginal oscillation in the subsequent epochs, peaking at the 30th epoch. Training loss starts at 1.75 and significantly decreases to 0.05 within a maximum of 15 epochs, indicating a well-trained model. Nonetheless, over the epochs, validation loss drops from about 1.5 at the 10th epoch but begins to fluctuate

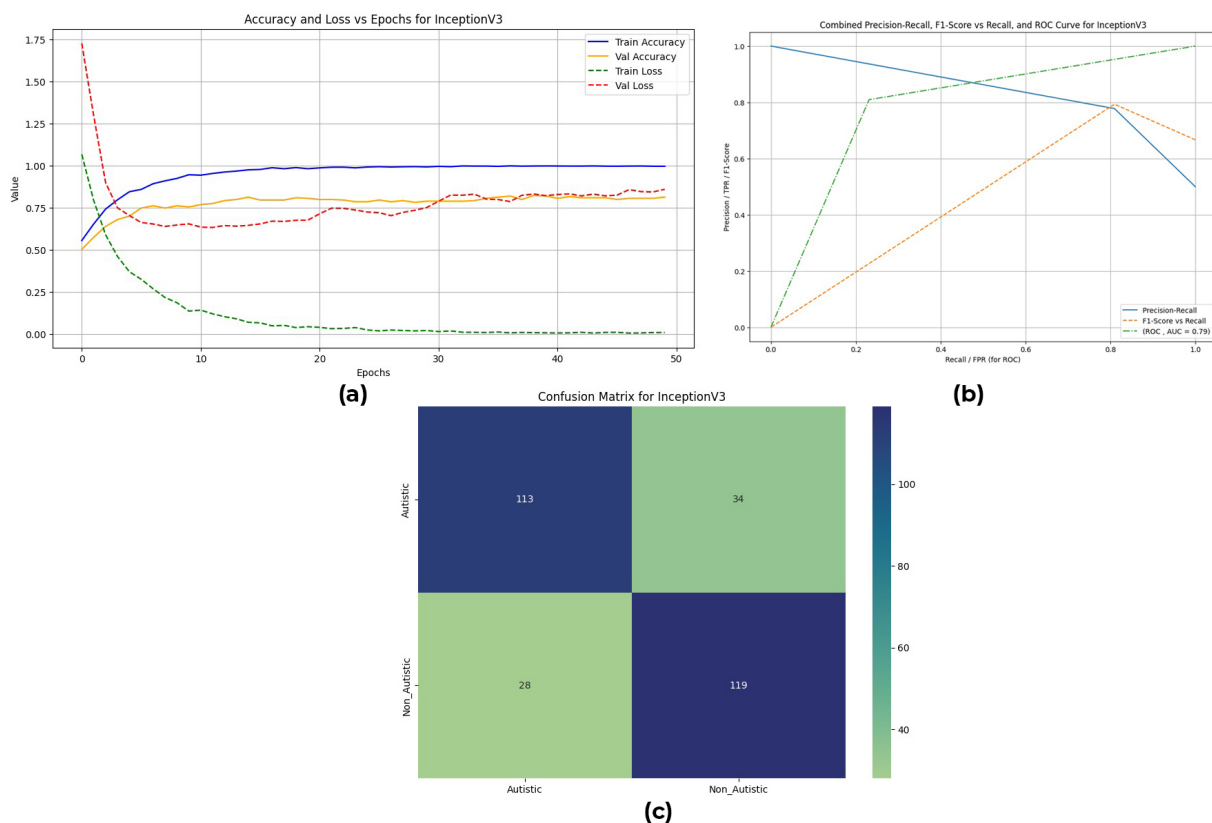


FIGURE 12. InceptionV3 Model Evaluation.

thereafter, potentially due to factors such as overfitting or issues with the training/test data split. In (b), it is observed that while precision increases to a maximum value of 1.0 when there are no true positives classified as negatives, precision steeply declines as recall increases, highlighting the trade-off between precision and true positives. The F1-Score peaks when recall is slightly above 0.8 before declining as recall rises, which indicates a decline in overall system performance. Thus, when clustering the Autistic and Non-Autistic classes, the R.O.C. curve shows moderate accuracy with an A.U.C. of 0.79, suggesting the need for further improvements to enhance classification accuracy. From the confusion matrix provided in (c), it is easy to evaluate the performance of the ResNet50 model in determining whether an image is Autistic or Non-Autistic. In this case, 121 out of 169 images were correctly classified as having autism (True Positives), while 129 out of 171 images were correctly classified as not having autism (True Negatives). However, the model misclassified 26 images of autistic individuals as non-autistic (False Negatives) and 18 images of non-autistic individuals as autistic (False Positives). Assessment measures can be derived from this matrix, specifically concerning the correct classifications indicated by the diagonal totals of 121 and 129. The evaluation suggests that the model classified non-autistic images with fewer errors compared to the percentage of errors observed in the classification of autistic images.

6.6 RESNET50

In FIGURE 13, (a) shows the graph illustrating the training and validation accuracy and loss of the ResNet50 model over 50 epochs. The blue line represents training accuracy, which gradually increases and reaches around 0.95, indicating that the model performs strongly on the training dataset. This graph features four metrics: the green plot denotes 'Training Accuracy,' the blue plot shows 'Cross Entropy Loss,' the yellow plot indicates 'Validation Accuracy,' and the red plot represents 'Training Cross Entropy Loss.' Validation accuracy varies unsystematically but approaches 0.82 after the initial growth, suggesting a slight variation between the training and validation sets. Training loss is initially high but rapidly converges to 0.05, which is desirable. In contrast, the overall validation loss reduces and stabilizes at 0.4, implying that the model does not overfit and that further optimization of parameters could yield better results. In (b), three critical performance metrics are displayed: the Precision-Recall (P-R) curve, the F-measure against Recall graph, and the ROC curve. The Precision-Recall graph demonstrates a trade-off; as the Recall value rises to 0.8, the Precision value hovers

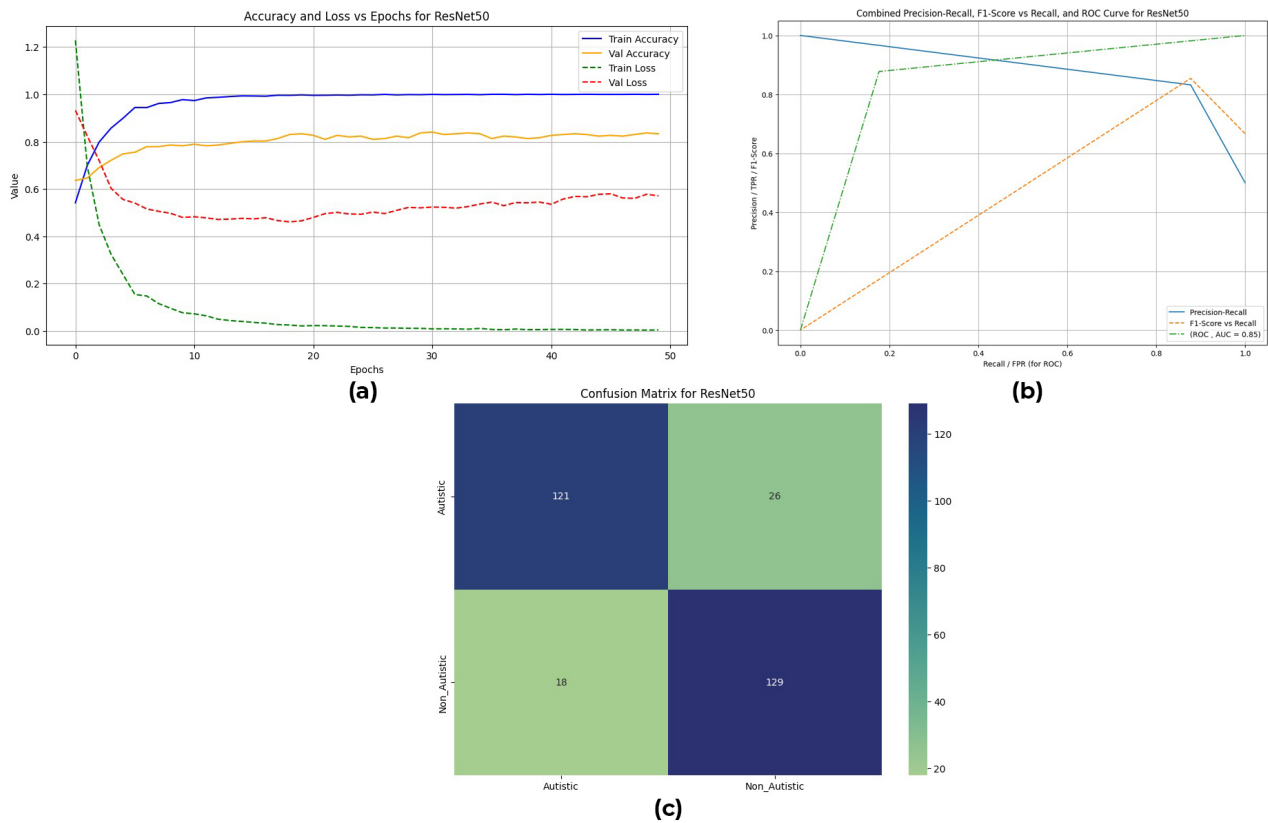


FIGURE 13. ResNet50 Model Evaluation.

around 0.6. On average, the F1-Score versus Recall graph indicates that the best balance of Recall and Precision occurs at a Recall of approximately 0.7, yielding an F1-score of around 0.85; however, the F1 score declines sharply below this point. The ROC curve, depicting the True Positive Rate (TPR) against the False Positive Rate (FPR), generates an AUC of 0.85, suggesting that the model has high classification capability to differentiate between classes. In the confusion matrix in (c), actual positive values denote the ResNet50's ability to classify images as Autism and Non-Autism. Out of a total of 169 images depicting Autism, the model accurately identified 121 as positive for Autism, yielding an actual positive rate of 0.71. Conversely, out of 171 non-autistic images, ResNet50 identified 129 correctly. Nevertheless, it misclassified 26 images of autistic individuals as non-autistic (False Negatives) and misidentified 18 non-autistic images as autistic (False Positives). Key performance measures can be derived from this matrix, particularly regarding diagnostic capacity, as indicated by the diagonal elements 121 and 129, which reflect a low error rate when the model is exposed to images of non-autistic individuals; however, the misclassification rate was higher among images of autistic individuals.

Table 2. Model Performance Comparison for Autism Detection

Model	Accuracy	Precision	Recall	AUC
VGG19	0.77	0.77	0.78	0.77
VGG16	0.78	0.78	0.77	0.78
XceptionNet	0.83	0.83	0.83	0.83
MobileNetV2	0.81	0.81	0.81	0.81
InceptionV3	0.79	0.79	0.79	0.79
ResNet50	0.85	0.85	0.85	0.85

Table 2 depicts the performance of multiple deep learning models for autism detection. The table consists of columns with the main evaluation criteria: accuracy, precision, recall, and AUC.

FIGURE 14 shows the result of the model used in the experiment to differentiate between people with autism and non-autism. Every face is described based on the classification to either be Autistic or Non Autistic and the confidence level

(for instance 0.99 indicates the certainty level). In the case of autistic subjects, the severity level (Mild, moderate, severe) is additionally predicted by the model. The green bounding boxes denote the correct faces detected by the algorithm.

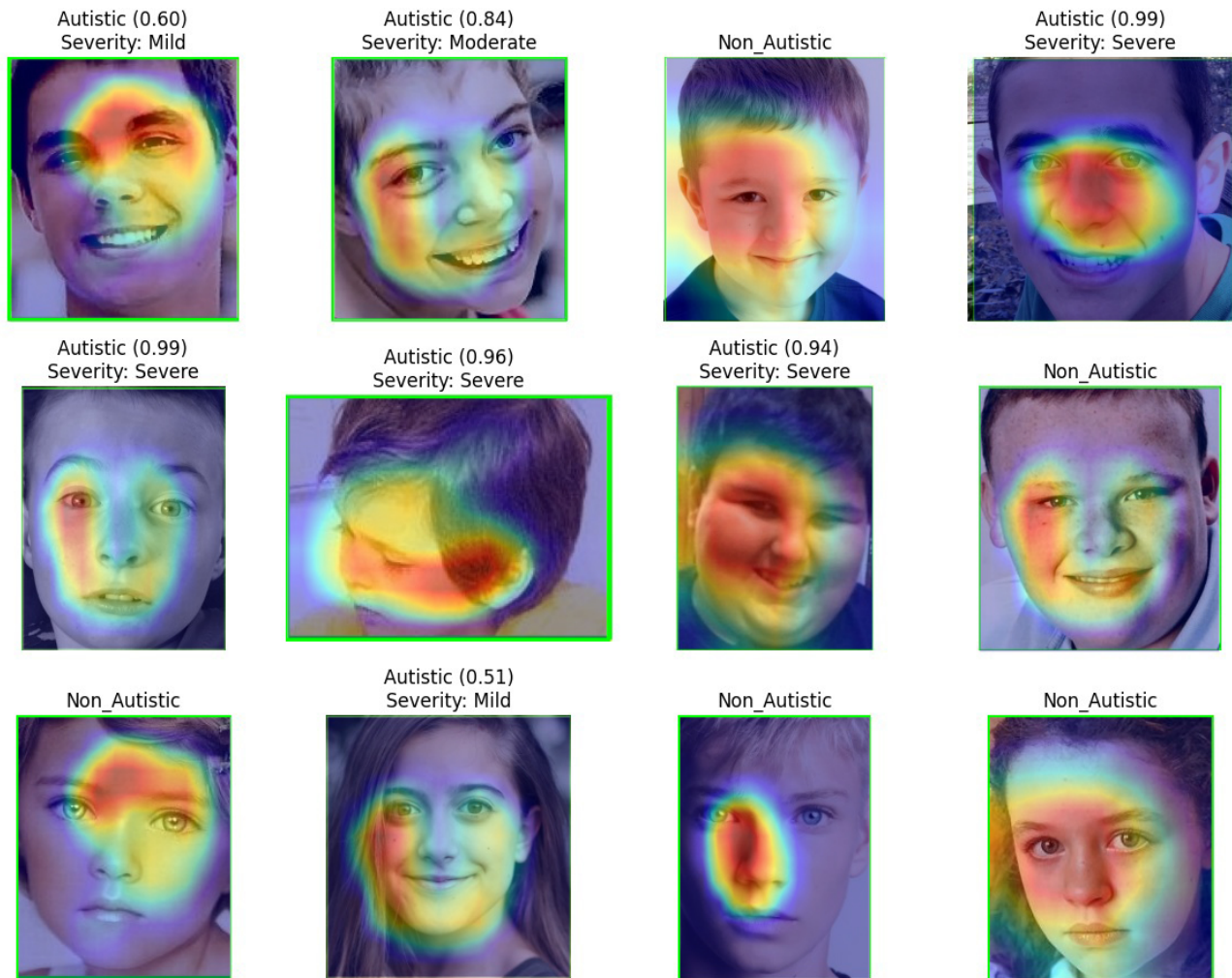


FIGURE 14. Results obtained from the test dataset for Autism detection with the best model successfully detected and classified on gradient-based between autistic and non-autistic individuals.

7. CONCLUSION

The study centered on the continuous spectrum analysis of autism spectrum disorder (ASD) employing six progressive deep learning models—VGG19, VGG16, XceptionNet, MobileNetV2, InceptionV3, and ResNet50—and analyzed the efficiency of the gradient-based classification models for ASD detection. To compare each model's performance, it was trained and tested using training, validation, and test datasets, and the accuracy, precision, recall and confusion matrix results captured each architecture's predictive accuracy and generalization. The high accuracy and detailed classification report reveal the possibility of using transfer learning and deep learning to diagnose ASD using facial image recognition. These findings show that using pre-trained models with the ASD dataset and fine-tuning helps distinguish between the autism and non-autism categories, which might create a template suitable for early intervention. From the six tested models, the differentiation in the architectural feature and training process affected the final test accuracy of different models, underlining the significance of the model choice for particular tasks in medical imaging. These findings also add to the enhancement of artificial intelligence-based diagnostic tools for identifying children with ASD.

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CONFLICT OF INTEREST

The authors declare no competing interests in the preparation of this manuscript.

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